JANMITRA

Civic Issue Reporting & Management System

Nagpur Municipal Corporation | 4 Categories | 4 Wards | 4 Departments

Team ResolveIT

Product Requirements Document (PRD)

What the product must do and for whom

CITIZEN PORTAL	Report Issue | Track Status | Verify Resolution
BACKEND (Django)	REST API | Auth | Business Logic | AI Module
AI VALIDATION	Fake Detection (EXIF+ELA+HuggingFace) | YOLO Relevance Check
DATABASE	PostgreSQL + PostGIS | 4 Categories | 4 Wards | 4 Depts
DEPT. DASHBOARD	PWD | Solid Waste | Water Works | Electrical

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1. Project Overview

JANMITRA is a web-based Civic Issue Reporting & Management System developed for Nagpur Municipal Corporation (NMC). Citizens report local civic problems through a simple mobile-friendly form. The system automatically classifies the complaint into one of 4 categories, maps it to the responsible NMC department, detects the ward, validates the photo using AI, assigns a worker from the matching department and ward, and provides real-time tracking to the citizen. Department officials manage their complaints through a dedicated department dashboard. Administrators have full system visibility.

Team	ResolveIT
Project	B.Tech Final Year Project -- Problem Statement SIH25031
Organisation	Nagpur Municipal Corporation (NMC)
Scope	4 Civic Categories | 4 NMC Departments | 4 Wards
Infrastructure	Free tools and hosting -- zero cost prototype

2. Problem Statement

Citizens in Nagpur who notice civic problems -- a pothole, overflowing garbage bin, broken streetlight, or blocked drain -- have no easy digital way to report them. Even when complaints are made (by phone or visiting offices), there is no tracking, no accountability, and no feedback to the citizen. Workers receive verbal orders with no priority or map data. Department officers have no real-time data on complaint volumes or resolution performance by ward.

- No structured digital complaint submission linked to GPS location and ward
- No automatic routing to the correct NMC department
- Fake or irrelevant complaint photos waste worker time -- no AI validation
- Workers have no priority-sorted task list or map navigation
- Citizens get zero updates after submitting -- no transparency
- Department heads have no real-time dashboard to monitor complaints by ward
- No data to identify chronic problem areas or underperforming departments

3. The Four Civic Categories

NOTE JANMITRA handles exactly FOUR civic issue categories, each mapped to a specific NMC department. No other categories exist. All complaints must fall into one of these four.

#	Category	Responsible Dept.	Organisation
1	Roads & Infrastructure	Public Works Department (PWD)	NMC
2	Garbage & Waste Management	Solid Waste Management Dept.	NMC
3	Water & Drainage	Public Health Engineering / Water Works Dept.	NMC
4	Street Lighting & Electrical	Electrical Department	NMC

Category Details

Roads & Infrastructure

Department: Public Works Department (PWD)

Examples: Potholes, damaged roads, broken footpaths, damaged dividers

Garbage & Waste Management

Department: Solid Waste Management Department

Examples: Uncollected garbage, overflowing bins, illegal dumping, dirty public areas

Water & Drainage

Department: Public Health Engineering / Water Works Dept.

Examples: Water leakage, no supply, dirty water, blocked drains, sewage overflow

Street Lighting & Electrical

Department: Electrical Department

Examples: Street light not working, damaged poles, dark streets

4. The Four Wards

The project covers 4 municipal wards. Every complaint is tagged to a ward. Every worker is assigned to one department and one ward. Complaint assignment follows the rule: worker's department must match complaint's category department AND worker's ward must match complaint's ward.

Ward	Coverage	Worker Assignment
Ward 1	NMC municipal zone 1	Workers linked to dept + Ward 1
Ward 2	NMC municipal zone 2	Workers linked to dept + Ward 2
Ward 3	NMC municipal zone 3	Workers linked to dept + Ward 3
Ward 4	NMC municipal zone 4	Workers linked to dept + Ward 4

5. User Roles

Role	Who	What They Can Do
Citizen	Any NMC area resident	Submit complaints, track status, verify resolution, rate closure
Worker	NMC field employee	See assigned tasks (dept+ward filtered), upload proof, update status
Dept. Officer	NMC department head	View dept. dashboard, validate complaints, assign workers in their dept.
Admin	NMC system admin	Full access: all departments, all wards, all workers, all analytics

6. Core Features

6.1 Complaint Submission

Field	Detail
Category	Dropdown with exactly 4 options (as above). Cannot proceed without selecting.
Ward	Dropdown with Ward 1, Ward 2, Ward 3, Ward 4. Citizen selects their ward.
Description	Text area. Minimum 20 characters. Clear description of the problem.
Photos	1 to 3 photos. Each photo is AI-validated before complaint is submitted.
GPS Location	Auto-detected from browser. Citizen can drag map pin to correct exact spot.
Complaint ID	System generates unique ID after successful submission. Shown to citizen.

6.2 AI Photo Validation (Two Layers)

Every photo goes through two AI checks before the complaint is accepted:

- Layer 1 -- Fake Detection: EXIF metadata check, Error Level Analysis (ELA), and HuggingFace pre-trained fake image detector. Rejects AI-generated images, screenshots, and downloaded images.
- Layer 2 -- Relevance Check: YOLOv8 model checks if the image contains a real-world civic issue scene. Low-confidence complaints go to the department officer's manual review queue.

6.3 Automatic Department Routing

Based on the category selected, the complaint is automatically routed:

Roads & Infrastructure	-- Public Works Department (PWD)
Garbage & Waste Management	-- Solid Waste Management Department
Water & Drainage	-- Public Health Engineering / Water Works Dept.
Street Lighting & Electrical	-- Electrical Department

6.4 Worker Assignment (Department + Ward Rule)

NOTE Workers are assigned to complaints only when BOTH conditions match: (1) Worker's department matches the complaint's category department. (2) Worker's ward matches the complaint's ward. Admin can override this rule in exceptional cases.

- Department officer sees available workers filtered by their department + the complaint's ward
- System highlights if no worker is available in that department+ward combination
- Admin can assign any worker to any complaint regardless of ward (with a flag/note)
- Worker sees ONLY complaints from their department + their ward in their task list

6.5 Worker Task Management

- Task list sorted by priority score (Critical at top)
- Each task shows: complaint photos, description, GPS map, category, ward
- Worker updates status: Accepted > In-Progress > Work Done
- Worker uploads at least 1 proof-of-work photo before marking Done -- enforced
- Citizen receives push notification at each status change

6.6 Citizen Verification and Closure

Citizen Action	System Response
Accepts resolution	Complaint closed. Citizen rates 1-5 stars.
Rejects resolution	Complaint reopened with rejection note. Worker notified.
No response in 72 hours	System auto-closes via scheduled background task.

6.7 Department-Wise Dashboard

A dedicated dashboard for each of the 4 NMC departments. Department officers see only their department's data. Admin sees all departments. Dashboard includes:

Dashboard Widget	Description
KPI Summary Cards	Total, Pending, In-Progress, Resolved, Rejected complaints
Average Resolution Time	How long complaints take to close (by dept, by ward)
Severity Distribution	Breakdown by priority level (Critical/High/Medium/Low)
Ward-wise Complaint Count	Bar chart: how many complaints per ward in this department
Complaint Trends	Line chart: complaints over time (daily/weekly/monthly)
Worker Workload	Table: each worker, their active tasks, avg. resolution time
High-Priority Open List	Table: all Critical and High priority open complaints
Filters	Filter by: Date range, Ward, Category, Status, Severity

7. Functional Requirements

ID	Requirement
FR-01	Citizens register with name, mobile number, ward (1-4), password
FR-02	Login works for all 4 roles (Citizen, Worker, Dept. Officer, Admin)
FR-03	Each role sees only their designated portal and data
FR-04	Complaint form has exactly 4 category options + ward selection (1-4)
FR-05	Each photo checked for EXIF metadata (Layer 1 fake detection)
FR-06	Each photo checked with ELA analysis (Layer 1 fake detection)
FR-07	Each photo checked with HuggingFace fake detector (Layer 1)
FR-08	YOLOv8 checks image relevance to civic issues (Layer 2)
FR-09	Complaint auto-routed to department based on selected category
FR-10	Worker assignment enforces dept. + ward match rule
FR-11	Admin can override dept+ward rule with a justification note
FR-12	Workers see only their dept + ward complaints in task list
FR-13	Priority score calculated from severity, safety, public impact, duplicates
FR-14	Duplicate detection: same category + same ward + GPS within 100m
FR-15	Worker must upload proof photo before marking Work Done
FR-16	Citizen can Accept/Reject resolution; 72h auto-close if no response
FR-17	Dept. officer dashboard shows all metrics for their department only
FR-18	Admin dashboard shows all departments with cross-dept comparison
FR-19	Dashboard filters: date, ward, category, status, severity
FR-20	Push notifications via Firebase at every status change
FR-21	Manual validation queue for AI low-confidence complaints

8. Non-Functional Requirements

ID	Requirement	Target
NFR-01	API response time	Under 500ms for all standard operations
NFR-02	AI processing (background)	Under 30 seconds -- user never waits
NFR-03	Mobile responsiveness	Works on Android Chrome, all mobile browsers
NFR-04	Duplicate detection accuracy	Within 100 metres, same ward, same category
NFR-05	Authentication security	JWT tokens, RBAC enforced on every API endpoint
NFR-06	Data integrity	Worker-complaint ward/dept mismatch prevented at DB level
NFR-07	Infrastructure cost	Rs. 0 -- all free/open-source tools and hosting
NFR-08	Availability	UptimeRobot prevents Render sleep during demo

9. Out of Scope

- No additional civic categories beyond the 4 defined above
- No Flutter/native mobile app -- responsive web only
- No payment gateway or fine collection
- No direct integration with NMC's existing ERP or legacy systems
- No real-time video or IoT sensor integration
- No SMS notifications -- browser push only via Firebase FCM
- No multi-city or multi-municipality support in this version